Non-surgical management and analgesia strategies for older adults with multiple rib fractures: A systematic review, meta-analysis, and practice management guideline from the Eastern Association for the Surgery of Trauma and the Chest Wall Injury Society

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BACKGROUND: Chest wall injury in older adults is a significant cause of morbidity and mortality. Optimal nonsurgical management strategies for these patients have not been fully defined regarding level of care, incentive spirometry (IS), noninvasive positive pressure ventilation (NIPPV), and the use of ketamine, epidural, and other locoregional approaches to analgesia.

METHODS: Relevant questions regarding older patients with significant chest wall injury with patient population(s), intervention(s), comparison(s), and appropriate selected outcomes were chosen. These focused on intensive care unit (ICU) admission, IS, NIPPV, and analgesia including ketamine, epidural analgesia, and locoregional nerve blocks. A systematic literature search and review were conducted, our data were analyzed qualitatively and quantitatively, and the quality of evidence was assessed per the Grading of Recommendations Assessment, Development, and Evaluation methodology. No funding was used.

RESULTS: Our literature review (PROSPERO 2020-CRD42020201241, MEDLINE, EMBASE, Cochrane, Web of Science, January 15, 2020) resulted in 151 studies. Intensive care unit admission was qualitatively not superior for any defined cohort other than by clinical assessment. Poor IS performance was associated with prolonged hospital length of stay, pulmonary complications, and unplanned ICU admission. Noninvasive positive pressure ventilation was associated with 85% reduction in odds of pneumonia ($p < 0.0001$) and 81% reduction in odds of mortality ($p = 0.03$) in suitable patients without risk of airway loss. Ketamine use demonstrated no significant reduction in pain score but a trend toward reduced opioid use. Epidural and other locoregional analgesia techniques did not affect pneumonia, length of mechanical ventilation, hospital length of stay, or mortality.

CONCLUSION: We do not recommend for or against routine ICU admission. We recommend use of IS to inform ICU status and conditionally recommend use of NIPPV in patients without risk of airway loss. We offer no recommendation for or against ketamine, epidural, or other locoregional analgesia. (*J Trauma Acute Care Surg.* 2023;94: 398–407. Copyright © 2022 Wolters Kluwer Health, Inc. All rights reserved.)

LEVEL OF EVIDENCE: Systematic Review/Meta-analysis; Level IV.

KEY WORDS: Rib fractures; elderly; pain management; incentive spirometry; noninvasive positive pressure ventilation.

Rib fractures occur in 10% of trauma patients and are associated with lung contusions, hemopneumothorax, and cardiac injuries; they are also common in the elderly.[1] Complications of multiple rib fractures, including increased pneumonia rates, ventilator dependence, intensive care unit (ICU) and hospital length of stay (LOS), and mortality, in adults 65 years or older were compared with younger patients with similar injury severity.[2,3] Mortality risk increases by 19%, and nosocomial infection risk increased by 29% with each additional rib fracture.[2] Because of continuous chest wall motion during respiration, ribs require significantly

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longer to heal than fractures that can be immobilized; disability prevalence approaches 40% 6 months after injury.[4]

Because of the impact of rib fractures on outcomes in older adults, we developed a practice management guideline for this population.[3,5] Specifically, selection criteria for ICU admission, the use of locoregional anesthesia, and noninvasive positive pressure ventilation (NIPPV) are not well defined. Similarly, the utility of incentive spirometry (IS) and both single and multimodal pain medications are unclear in older adults. Recovery of patients from chest wall injury is prolonged, with low-severity injuries leading to chronic pain syndromes and high-severity injuries impacting quality of life and functional independence.[5] We performed a systematic review and meta-analysis concerning nonoperative management of chest wall injury.

PATIENTS AND METHODS

Objectives

A joint working group of Eastern Association for the Surgery of Trauma, Chest Wall Injury Society, and American Association for the Surgery of Trauma Patient Assessment Committee members evaluated nonoperative interventions for adults 65 years or older presenting with multiple rib fractures. Grading of Recommendations Assessment, Development, and Evaluation methodology was used as outlined by Kerwin et al.[6]

Selection of Outcomes and PICO Questions

Candidate outcomes for each PIC stem were voted upon by each author using a 1 to 9 scale. Outcomes scoring 7 to 9 were considered critical and included in our analysis. The outcomes and scores are summarized in Supplementary Table 1, http://links.lww.com/TA/C769. The following Population(s), Intervention(s), Comparison(s), and Outcome (PICO) questions were formulated before literature search:

1. In adults 65 years or older with three or more rib fractures (P), should admission to an ICU setting (I) versus admission to a non-ICU setting (C) take place to reduce pneumonia, need for intubation, ventilator days, or mortality (O)?
2. In adults 65 years or older with rib fractures (P), should routine use of IS (I) versus no routine use of IS (C), be performed to reduce pneumonia, need for intubation, or mortality (O)?
3. In adults 65 years or older with rib fractures and acute hypoxic respiratory failure refractory to nasal cannula and face mask (P), should NIPPV (high-flow nasal cannula [HFNC], bilevel positive airway pressure [BiPAP], continuous positive airway pressure [CPAP]) (I) versus endotracheal intubation (ETT) (C) be used to reduce pneumonia, need for intubation, ventilator days, or mortality (O)?
4. In adults 65 years or older with three or more rib fractures and dyspnea or refractory pain (P), should ketamine infusion plus structured multimodal pain therapy per institutional protocol (I) versus structured multimodal pain therapy per institutional protocol alone (C) be performed to reduce pain, pneumonia, need for intubation, ventilator days, or mortality (O)?
5. In adults 65 years or older with three or more rib fractures and dyspnea or refractory pain, should a thoracic epidural catheter and structured multimodal pain therapy per institutional protocol (I) versus structured multimodal pain therapy per institutional protocol alone (C) be performed to reduce pain, pneumonia, need for intubation, hospital LOS, or mortality (O)?
6. In adults aged 65 years or older with three or more rib fractures and dyspnea or refractory pain, should nonepidural locoregional anesthetic (subcutaneous infusion pump or local block) and structured multimodal pain therapy per institutional protocol (I) versus structured multimodal pain therapy per institutional protocol alone (C) be performed to reduce pain, pneumonia, need for intubation, ventilator days, or mortality (O)?

Literature Search

This study conforms with the PRISMA guidelines, and a complete checklist has been uploaded as Supplementary Table 2, http://links.lww.com/TA/C770, along with the abstract checklist as Supplementary Table 3, http://links.lww.com/TA/C771. Our systematic review was registered with PROSPERO (2020-CRD42020201241). Search strategy included MEDLINE, EMBASE, Cochrane Library, and Web of Science (Supplementary Table 4, http://links.lww.com/TA/C774). Published retrospective case-control, prospective observational, and randomized studies in English that included both patients receiving the intervention and comparator were eligible for review. Case series and reports, commentaries, animal studies, and operative technique articles were excluded. Systematic reviews were assessed to ensure that their referenced primary studies were included.

Review of Abstracts and Full-Text Articles

Titles, abstracts, and full texts were reviewed independently by two team members, and any conflicts were adjudicated by a third. All reviews took place in Covidence (www.covidence.org). The list of studies (Supplementary Table 5, http://links.lww.com/TA/C773) and PRISMA flow diagram (Fig. 1) are included.

Data Extraction and Analysis

Data extraction was performed by a single author (K.M.) with group oversight and review to ensure accuracy. Meta-analysis was performed using RevMan 5.0 (Cochrane, Inc., London, UK, http://revman.cochrane.org), including odds ratios (ORs) and 95% confidence intervals. Heterogeneity was calculated and quantified (I^2 < 50%, acceptable; 50–74%, borderline; ≥75%, high).[7]

Assessment of Quality of Evidence

Quality of evidence was assessed with GRADEPro (www.gradepro.org), considering potential risk of bias, inconsistency, indirectness, imprecision, and publication bias. Authors voted on recommendations for each PICO question, evaluating the relationship of benefits and harms, patient values and preferences, and resource utilization.[6]

RESULTS

1: ICU Admission

Qualitative Analysis

Six retrospective studies were included. Athanassiadi et al.[8] found that Injury Severity Score (ISS) predicted morbidity and hospital LOS but did not discuss ICU admission. Bellone et al.[9] found that trauma score and obliged orthopnea were associated with ICU admission within the first 72 hours. Blecher et al.[10]

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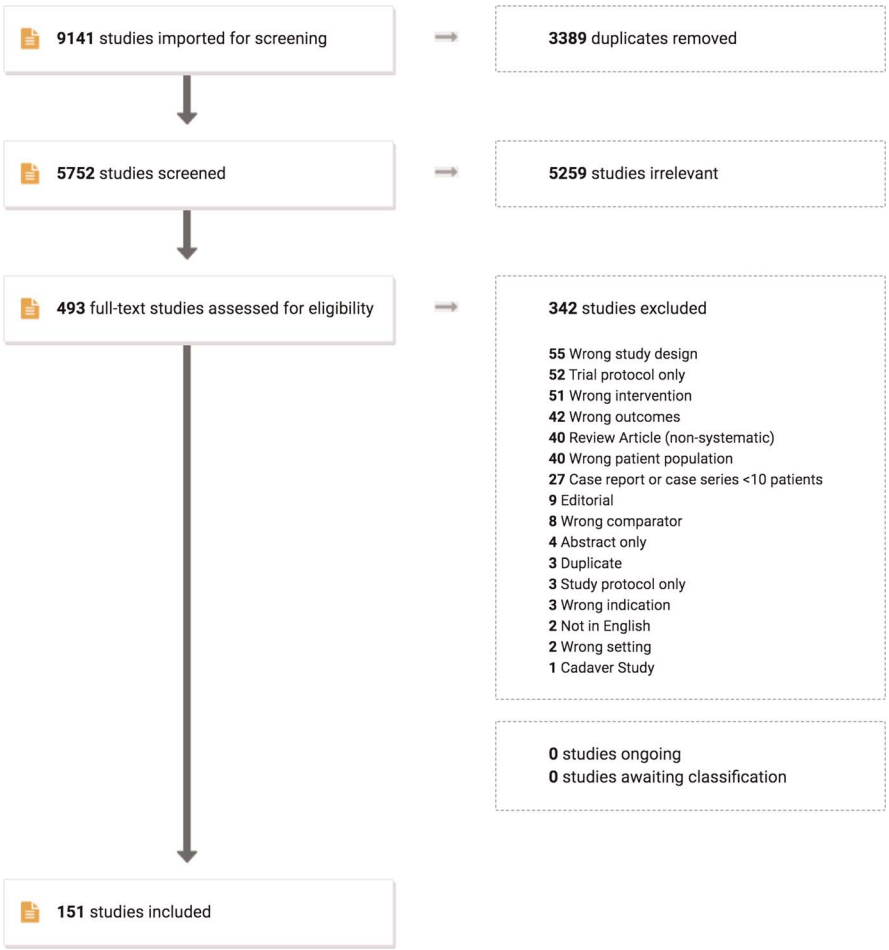


Figure 1. The PRISMA flow diagram indicates the flow of studies and their review process for the systematic review.

found that nearly 10% (70 of 764) of patients initially triaged to the floor required ICU admission, and that tube thoracostomy was an independent predictor of ICU need. Bowman et al.[11] found that IS <1 L, use of a walker, chest Abbreviated Injury Scale (AIS) score of 3 or higher, age 72 years or older, and active smoking were independent risk factors for critical patient incidents in non-ICU patients, although there were only 7 incidents in the 150-patient study. Pyke et al.[12] studied outcomes before and after implementation of a protocol admitting older patients with three or more rib fractures to the ICU. Although the recommendation was not universally followed, the postintervention group had fewer complications, unplanned ICU admissions, and nonhome discharges.[12] Shi et al.[13] found, conversely, when their institutional guideline for ICU admission for patients 65 years or older with two or more rib fractures was not followed, only 9% of patients had delayed ICU admission.

These studies were retrospective with different definitions. Some studies included older patients,[11–13] while other studies included all patients. There was little discussion regarding the predetermined critical outcomes. Guidelines for ICU admission were also frequently not followed. Risk of bias was high, and quality of evidence was very low. These data did not support any quantitative analysis nor any recommendations.

2: Incentive Spirometry

Qualitative Analysis

There were three retrospective studies and one randomized clinical trial (RCT) using IS. Bakhos et al.[14] noted that vital capacity (actual and percent predicted), related to IS, correlated with hospital LOS and discharge home. The study population had median age of 80 years with 3.6 ± 1.6 rib fractures. Patients with a vital capacity of <1.4 L or <55% of predicted had a hospital LOS of more than 3 days.[14] Kelley et al.[15] used IS to guide treatment with NIPPV as part of a chest injury protocol that reduced pneumonia, unplanned intubation, and unplanned ICU admission in 316 patients. If IS was <750 mL despite the NIPPV trial, the settings were increased, and additional pain medications were added with consideration for ICU transfer.[15] In a retrospective review, Sadler et al.[16] noted that patients with IS of <1 L had a 3.3-fold increased risk of pulmonary complications and increased likelihood of nonhome discharge. In this study, the mean age was 54 years, and patients had an average of four rib fractures.[16] In the 50-patient RCT, patients receiving IS were less likely to have pulmonary complications, largely defined by retained hemothorax. Pulmonary function testing was also performed and showed decreased forced vital capacity and forced expiratory volume in

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1 second in the control group. The population in this group had a mean age of 55 years, and approximately 80% of patients had at least three rib fractures.[17]

In assessing the literature, data quality was low with one RCT and three retrospective studies, and no quantitative analysis was possible. However, other factors considered included good therapeutic tolerance, few adverse effects, and low cost. Thus, we recommended that IS be used to reduce overall pulmonary complications.

PICO 3: NIPPV

Qualitative Analysis

This field of literature included two RCTs directly addressing the topic and a third with a different control group, as well as three retrospective studies. Carrie et al.[18] studied NIPPV as part of an overall blunt chest trauma protocol with 1:1 nearest neighbor matching. The use of NIPPV increased from 6% to 39% with the protocol, but no results were reported directly attributable to the use of NIPPV.[18] Noninvasive positive pressure ventilation was likewise part of the protocol developed by Kelley et al.,[15] which noted a reduced rate of unplanned ICU admissions, unplanned intubations, and pneumonia; it is again unclear exactly what proportion of these outcomes is attributable to NIPPV. Halub et al.[19] conducted a retrospective study of HFNC set to 50 L/min at 50% FIO_2 in patients with median age of 63 years and ISS of 21. They reported an 18% failure rate for HFNC, with 69% of patients on HFNC never needing intubation.[19]

Hernandez et al.[20] performed a single-center RCT of patients with mean age in the mid-40s, mean chest AIS score of 4, and PaO_2/FIO_2 ratio of <200 for more than 8 hours within the first 48 hours after injury. The control arm received supplemental oxygen (≥10 L/min) by mask, and the experimental arm received NIPPV with a BiPAP mask set to 10 to 12 cm H_2O inspiratory positive airway pressure and 6 cm H_2O expiratory positive airway pressure. The intubation rate was 40% in the control arm versus 12% in the NIPPV arm ($p = 0.02$).[20] This study could not be included in quantitative analysis, as the control group did not include ETT. However, it does provide good quality evidence that NIPPV may be beneficial and more closely approximates the practice pattern at many institutions.

The RCT by Bolliger and Van Eeden[21] compared patients older than 40 years with more than three rib fractures. Patients were randomized to CPAP mask and epidural analgesia or ETT. The CPAP group demonstrated a shorter treatment course (4.5 days vs. 7.3 days, $p = 0.0003$), shorter ICU and hospital LOS, and a lower pneumonia rate.[21] The RCT by Gunduz et al.[22] compared patients in their 30s to 40s with five or more consecutive rib fractures or flail chest, respiratory rate of >25 (resp/min), O_2 saturation <90% on 10 L face mask, and a PaO_2/FIO_2 ratio of <300 on ≥50% FIO_2 in the ICU. Twenty-seven patients received mechanical ventilation, and 25 received CPAP. The CPAP group had a lower rate of nosocomial infection and higher survival.[22] Overall quality of evidence was moderate, with some indirectness in multiple studies.

Quantitative Analysis

The studies by Bolliger and Van Eeden,[21] and Gunduz et al.[22] were also suitable for meta-analysis for two endpoints: pneumonia and mortality (Fig. 2*A* and *B*) with a total of 58 patients in the NIPPV group and 54 in the ETT group. The pooled odds ratio (OR) for pneumonia was 0.15 (95% CI, 0.06, 0.39) with $p < 0.0001$ in favor of CPAP. The pooled OR for mortality was 0.19 (95% CI, 0.04, 0.86) with $p < 0.03$ in favor of CPAP. The heterogeneity was 0% in both comparisons. Data from the meta-analysis were favorable with low risk of bias. However, the group felt that a full recommendation for NIPPV should not be offered because all the studies had multiple exclusions, including patients with traumatic brain injury, GCS score of <15, or inadequate airway protection. This was done to ensure that patients would not suffer adverse events due to aspiration or other airway events. Thus, we suggest that NIPPV be used in carefully selected patients with persistent acute hypoxic respiratory failure as defined as O_2 saturation <90% on supplemental oxygen ≥10 L by face mask after optimizing analgesia and ensuring that patients could protect their airway and avoid complications of airway failure. Patients whose O_2 saturations do not improve higher than 90% on NIPPV or who have worsening of their mental status or airway compromise should then proceed to ETT.

PICO 4: Ketamine Infusion

Qualitative Analysis

The literature concerning the use of ketamine for analgesia in adults with rib fractures consists of a retrospective case-control study of 30 patients and 2 RCTs that were organized similarly but included 2 differing populations. The retrospective study by Walters et al.[24] of ketamine infusion at 0.1 mg/kg/h for ICU patients (mean age, 59 years) with one or more rib fractures and ISS of >15 demonstrated that numeric pain score and opioid use were lower in the ketamine group compared with the standard group with a hydromorphone patient-controlled analgesia pump. The study by Carver et al.[25] included adults 64 years or younger with three or more rib fractures and randomized patients to ketamine at 0.15 mg/kg/h versus normal saline. This study found no significant difference in numeric pain score or oral morphine equivalent (OME) use after 24 hours in the overall patient population but did find a reduction in OME for patients with ISS >15 in the first 24 and 48 hours and overall.[25] A similar trial conducted by Kugler et al.[23] of adults 65 years or older with three or more rib fractures randomized patients to ketamine at 0.12 mg/kg/h versus normal saline. Both the Kugler et al.[23] and Carver et al.[25] studies used an institutional thoracic pain management protocol using both opioid and nonopioid medications; further details were not provided. This study found no significant effect of ketamine in the overall population but did note a decrease in OME for the ketamine group in the first 24 hours.[23]

Quantitative Analysis

The trials by Kugler et al.[23] and Carver et al.[25] were amenable for meta-analysis for numeric pain score but demonstrated no significant treatment effect (OR, 0.24 [95% CI, −0.64, 1.12]; $p = 0.59$; Fig. 2*C*). Overall, there were other concerns with the published literature, including the lack of standardization of pain management in the control groups and the indirectness associated with the Carver et al.[25] trial from the exclusion of older patients. Quality of evidence was felt to be low to moderate. No recommendation for or against the use of ketamine was indicated.

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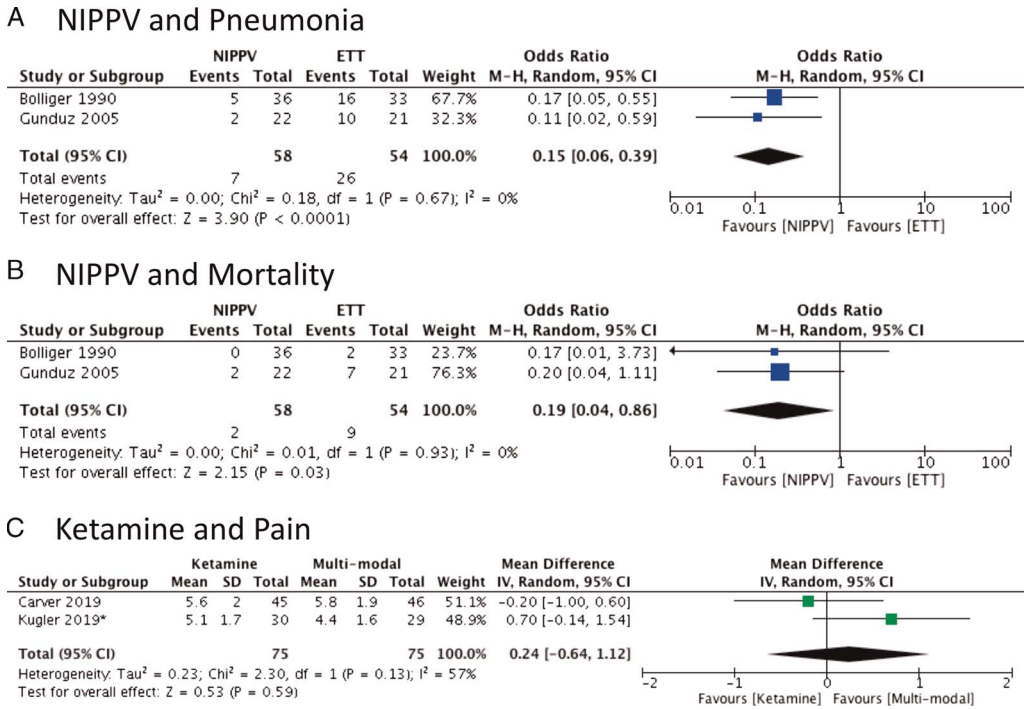


Figure 2. (*A*) Noninvasive positive pressure ventilation is associated with decreased incidence of pneumonia in two randomized controlled clinical trials. (*B*) Noninvasive positive pressure ventilation is associated with decreased incidence of pneumonia in two randomized controlled clinical trials. (*C*) Ketamine infusion is not demonstrably superior to multimodal pain medication in two closely related randomized controlled clinical trials. The trial by Kugler et al.[23] is asterisked, as it specifically included elderly patients.

PICO 5: Epidural Analgesia

Qualitative Analysis

The literature had several studies that could not be used for further analysis. Six studies were prospective or randomized comparisons of either different epidural techniques or epidural and locoregional techniques with no other control group.[26–31] Five studies were either case series or evaluated care bundles.[18,32–35] There were 12 retrospective studies with 3,916 patients (913 with epidural catheters and 3,903 without) with very low to low quality of evidence.[2,36–46] There were also 4 RCTs with a total of 79 patients, half with epidurals and half without. The study by Ahmed et al.[47] enrolled only mechanically ventilated patients between 18 and 55 years with three or more rib fractures and a flail segment. Bulger et al.[48] enrolled patients with three or more rib fractures and a mean age in the 40s. Mackersie et al.[49] included patients with three or more rib fractures, flail chest or sternal fracture, or two or more rib fractures and exploratory laparotomy or pulmonary contusion, with a mean patient age in the 40s. Ullman et al.[50] included patients with mean ages in the 40s to 50s, multiple rib and/or pelvic fractures, and ICU admission.

Among the retrospective and prospective studies, a surprisingly small number actually studied pain as an outcome. Furthermore, there was no standardization of the pain management in the control arm, with most studies using a mixture of intravenous and oral opioids and little use of nonopioid analgesics. For example, the study by Bulger et al.[2] indicated that 38% of patients older than of 65 years received acetaminophen and 14% received ketorolac; use of GABA-ergic agents was not tabulated. Essentially all older patients received some combination of intravenous and oral opioids.[2] There was one retrospective study demonstrating some improvement in numeric pain scores when epidural bupivacaine and fentanyl were compared with intravenous patient-controlled analgesia with intravenous morphine, but this was not corroborated in the single prospective study available, which compared epidural versus intravenous fentanyl administration.[44,49] An additional retrospective study that focused on delirium rates did not find a change in numeric pain scores but did note a decreased daily morphine equivalent dose after epidural catheter placement.[42] Because of the number of studies and the differences in study methodology, we proceeded directly to quantitative analysis for the other outcomes.

Quantitative Analysis

Among retrospective studies, use of epidural analgesia was associated with increased risk of ETT (OR, 1.71 [1.17–2.51]; $p = 0.006$; Fig. 3*A*), in data with relatively low heterogeneity. There was no significant relationship between epidural use and either pneumonia (Fig. 3*B*) or mortality (Fig. 3*C*); data had acceptable heterogeneity ($I^2 = 30\%$). Epidural use was associated with a prolonged hospital LOS (2.47 days [1.28–3.65 days]; $p < 0.0001$; Fig. 3*D*) and did not have a discernible effect on length of mechanical ventilation (Supplementary Fig. 1*A*, http://links.lww.com/TA/C780). The overall quality of evidence was very low to low.

Among prospective studies with epidurals, two studies demonstrated a nonsignificant trend toward reduced pneumonia (OR, 0.37 [0.12–1.15]; $p = 0.09$; Fig. 4*A*). There was no significant effect on mortality, hospital LOS, or length of mechanical ventilation, and these data had high heterogeneity (Fig. 4*B* and *D*). Quality of evidence was moderate rather than high because of the absence of standardization among pain control regimens. Eleven

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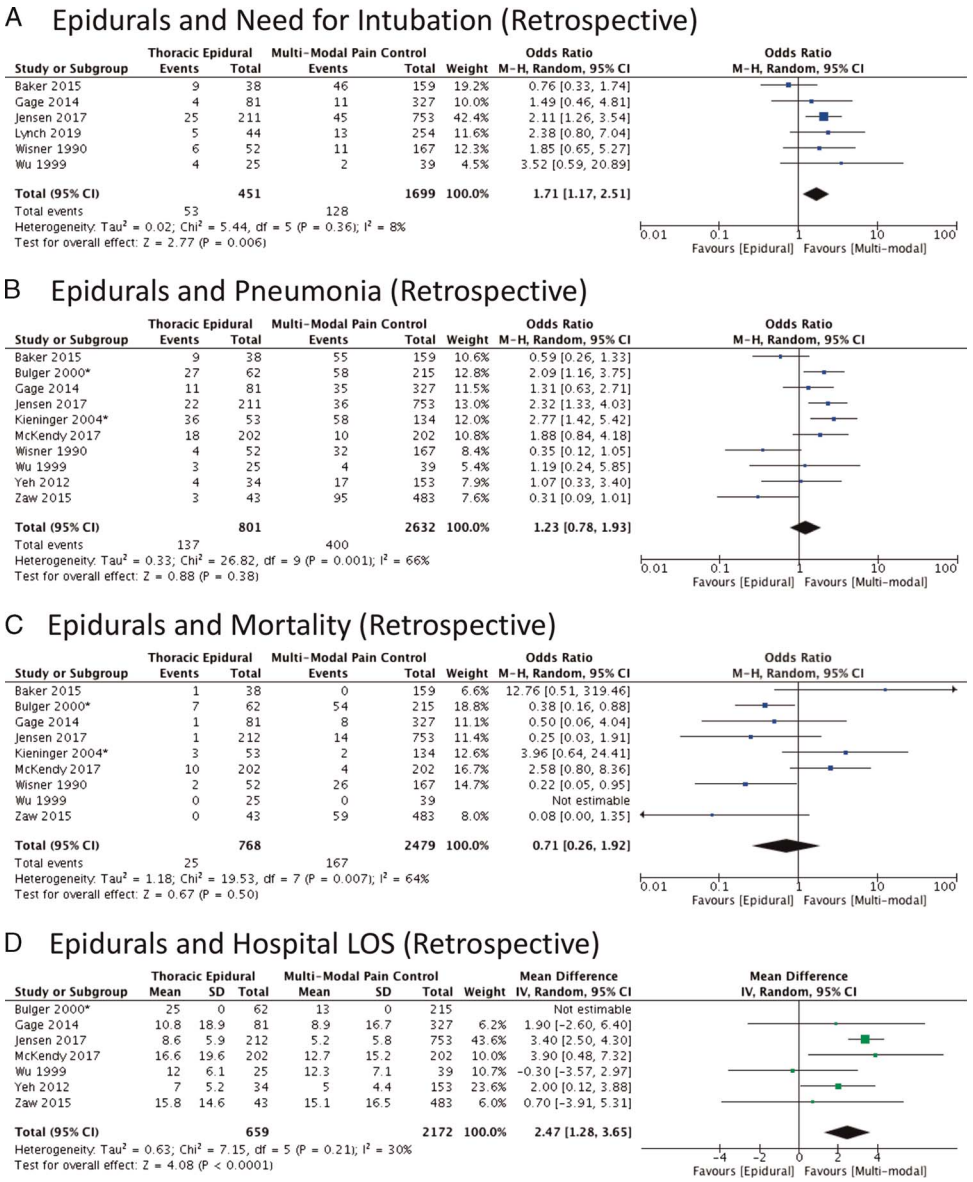


Figure 3. (*A*) Retrospective studies evaluating the need for ETT and mechanical ventilation with and without epidural analgesia are listed. There is a net effect that favors treatment without epidural analgesia. (*B*) Retrospective studies are evaluated for the incidence of pneumonia with and without epidural analgesia. There is no net effect. The study by Kieninger et al.[39] is asterisked, as it specifically included elderly patients. (*C*) There is no net effect on mortality with the use of epidural analgesia. The study by Kieninger et al.[39] is asterisked, as it specifically included elderly patients. (*D*) Retrospectively, epidural analgesia is associated with increased hospital LOS in days. The study by Bulger et al.,[48] is asterisked as it specifically included elderly patients.

of 19 coauthors favored no recommendation for or against, four favored a suggestion for epidural use, three favored against, and one voted to recommend against. The group therefore elected to make no recommendation for or against epidural analgesia.

PICO 6: Nonepidural Locoregional Analgesia

Qualitative Analysis

The field of literature had several studies that could not be used for further analysis. Seven were case series, two of which evaluated pain before and after locoregional analgesia.[51–57] The study by Hernandez et al.[56] noted a decrease in numeric pain score from a median of 7 before serratus anterior block to a median of 3, 4 hours after the block, with the type of local anesthetic at provider discretion. No subsequent pain scores were obtained. Adhikary et al.[57] indicated that use of an erector spinae block with ropivacaine reduced pain scores from a mean of 8 (24 hours prior to block) to 5 to 6 in the next 24 hours. Six studies were comparisons primarily between epidural and other locoregional techniques, without a standardized control group.[26,27,29–31,40] One study evaluated a care bundle, while another predominantly studied delirium and did not break down outcomes by type of locoregional analgesia.[18,42] This left two RCTs. Ge et al.[58] studied 90 patients with a mean age of 40 years and an average of 4 rib

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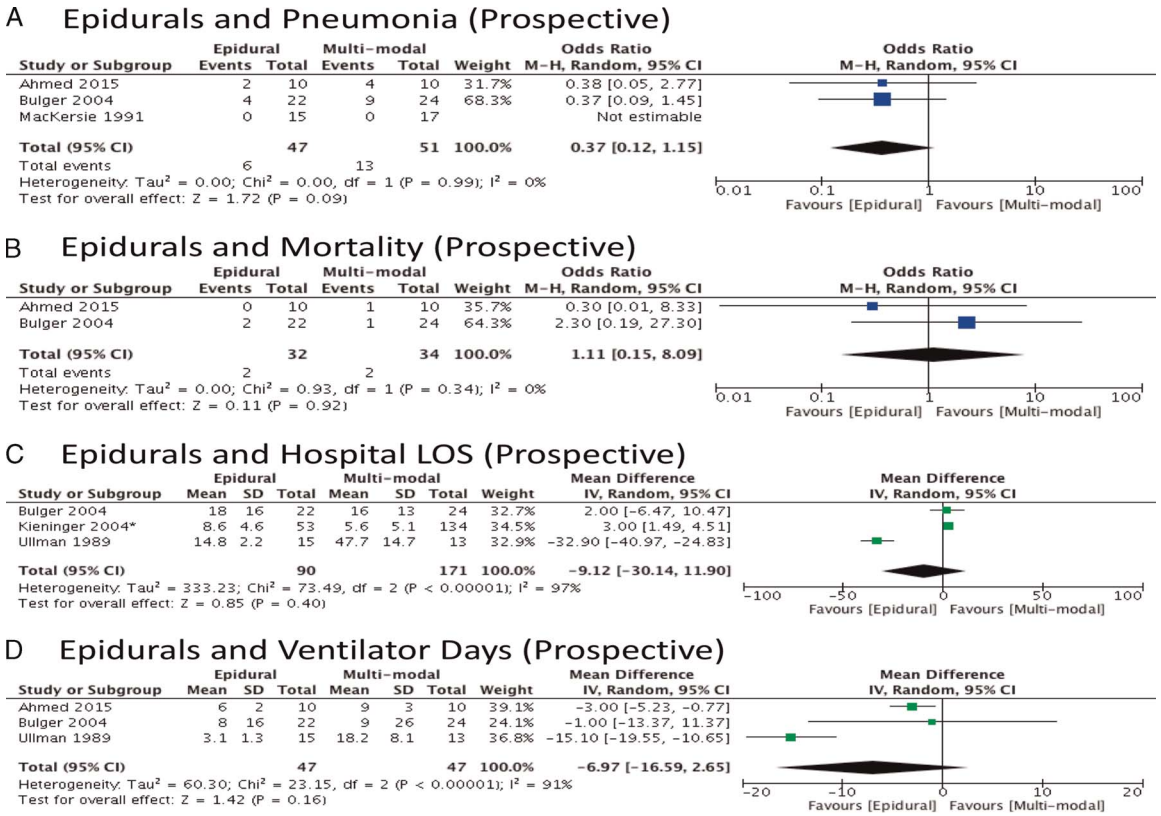


Figure 4. (*A*) Prospective studies do not indicate a benefit for epidural analgesia with respect to pneumonia. (*B*) Prospective studies do not indicate a benefit for epidural analgesia with respect to mortality. (*C*) Hospital LOS in days is not significantly different with or without epidural analgesia. The study by Kieninger et al.[39] is asterisked, as it specifically included elderly patients. (*D*) Epidural analgesia does not result in a significant difference in ventilator days.

fractures and randomized patients to intravenous opioid patient controlled analgesia versus thoracic paravertebral catheters. This study reported reduced numeric pain scores and improved pulmonary function tests in the paravertebral catheter group. Visual analog pain scores at rest decreased by 1 point on day 1, 1.1 points with coughing on day 1, 0.7 points at rest on day 2, 0.6 points with coughing on day 2, 0.2 points with coughing on day 3, and 0.1 points with coughing on day 4.[58] Gabram et al.[59] conducted an RCT of 42 patients in their 50s with approximately 4 rib fractures and a forced vital capacity <70% of predicted. There were baseline differences between groups; the intrapleural catheter group had fewer rib fractures and a lower initial percentage of predicted forced vital capacity. There were no significant differences in the results.[59] The overall quality of these studies was moderate because of potential allocation differences in the Gabram et al.[59] study and absence of nonopioid analgesics.

Quantitative Analysis

The studies performed by Gabram et al.[59] and Ge et al.[58] were amenable to meta-analysis for pneumonia (Supplementary Fig. 1*B*, http://links.lww.com/TA/C780). There was no difference identified, and heterogeneity was acceptable (I^2 = 44%). Twelve of 19 coauthors voted for no recommendation, five favored a suggestion, one voted for a recommendation, and one favored a suggestion against. The final decision was to offer no recommendation for or against.

DISCUSSION

Grading the Evidence

For PICO 1 (ICU), the included studies were all retrospective in nature and also indirect, assessing either outcomes or a larger population than the PICO question. There were also varying inclusion criteria based on injury severity, resulting in very low quality of evidence (Supplementary Table 6, http://links.lww.com/TA/C774). For PICO 2 (IS), there were three retrospective studies and one RCT, with overall low quality based on nonserious indirectness (one study used assessment of vital capacity rather than IS). For this PICO question, the mean age was in the mid-50s with at least three rib fractures in most patients (Supplementary Table 7, http://links.lww.com/TA/C775). For PICO 3 (NIPPV), the quality of evidence began as very high because of the inclusion of two RCTs with large effect sizes but was downgraded 1 point for indirectness because of age and injury severity variance between the RCTs and the PICO population (Supplementary Table 8, http://links.lww.com/TA/C776). For PICO 4 (ketamine), the quality of evidence began as high because of two RCTs with small effect size but was then downgraded twice: first for risk of bias due to variations in pain management techniques in the control group and absence of nonopioid medications, and second for indirectness because one RCT explicitly excluded older patients (Supplementary Table 9, http://links.lww.com/TA/C777). For PICO 5 and 6 (epidurals and other locoregional anesthesia), the quality of evidence was

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TABLE 1. Summary of the recommendations for all PICOs

PICO	Recommendations	Brief Rationale	Voting
PICO 1 ICU admission	In adults older than 65 y with ≥3 rib fractures, *no recommendation* is made for or against ICU versus non-ICU admission.	– Insufficient evidence	Unanimous
PICO 2 IS	In adults older than 65 y with rib fractures, *we recommend* routine use of IS.	– Moderate evidence for benefit and risk stratification – Few adverse effects – Inexpensive	Unanimous
PICO 3 NIPPV	In adults older than 65 y with rib fractures and acute hypoxic respiratory failure refractory to nasal cannula and face mask, *we suggest* NIPPV be attempted in suitable candidates (in which intubation is not needed for airway protection).	– Strong evidence for risk reduction for pneumonia and mortality – Concern for potential for airway compromise	Unanimous
PICO 4 Ketamine	In adults older than 65 y with ≥3 rib fractures and dyspnea or refractory pain, *no recommendation* is made for or against ketamine infusion versus structured multimodal pain therapy per institutional protocol.	– Insufficient evidence for superior pain control – No benefit to other outcomes	Unanimous
PICO 5 Thoracic epidural catheters	In adults older than 65 y with ≥3 rib fractures and dyspnea or refractory pain, *no recommendation* is made for or against thoracic epidural versus structured multimodal pain therapy per institutional protocol.	– Insufficient evidence for superior pain control – No benefit to other outcomes	11 of 19 no recommendation 4 of 19 suggest 3 of 19 suggest against 1 recommend against
PICO 6 Nonepidural locoregional anesthesia	In adults older than 65 y with ≥3 rib fractures and dyspnea or refractory pain, *no recommendation* is made for or against nonepidural locoregional anesthetic versus structured multimodal pain therapy per institutional protocol.	– Insufficient evidence for superior pain control – No benefit to other outcomes	12 of 19 no recommendation 5 of 19 suggest 1 of 19 suggest against 1 of 19 recommend

initially high because of RCTs but then downgraded three times: first for serious risk of bias due to variations in the control pain regimen between the studies, second for inconsistency due to heterogeneity, and, finally, for indirectness due to age variance between the study populations and the PICO population, resulting in a final classification of very low to low quality of evidence (Supplementary Table 10, http://links.lww.com/TA/C778, and Supplementary Table 11, http://links.lww.com/TA/C779).

Limitations

This systematic review has some limitations. Focusing on patients 65 years or older led to a large amount of available literature being indirect and thus lowered the quality of applicable evidence. Literature in languages other than English was excluded, while some retrospective studies were included because of the low number of high-quality RCT in this area. Variation in trial methodologies and populations occurred frequently. Finally, in any meta-analysis, the possibility of differing results and interpretations thereof in the field of literature elicited always exists, which has the potential to alter study conclusions.

Recommendations

Recommendations are summarized in Table 1.

Using These Guidelines in Clinical Practice

This practice management guideline highlights the evidence pertaining to the nonsurgical management of patients 65 years or older with rib fractures. Although we cannot offer a recommendation on ICU admission for all patients older than 65 years, ICU admission should be considered in patients who also have frailty or deconditioning, hypoxemia (room air O_2 saturation <92%), IS <1 L, more severe chest injuries (chest AIS score, >3), use of a walker, and smoking.[11,13,60–62]

Incentive spirometry and effective pain control are mainstays of management. Poor use of IS (<1 L) or presence of tube thoracostomy should prompt close monitoring.[10,16] Noninvasive positive pressure ventilation (including HFNC and BiPAP) is suggested for selected patients 65 years or older with rib fractures and acute hypoxic respiratory failure refractory to nasal canula and face mask who do not require airway protection for other reasons (hemodynamic instability, traumatic brain injury, somnolence, severe acidosis, etc.) and have no contraindication for face mask use (agitated or uncooperative, multiple facial fractures, abdominal distension, etc.).[20–22] Noninvasive positive pressure ventilation failure should result in prompt ETT.

Adequate pain control is essential in the treatment of rib fractures. Data are lacking to support the use of ketamine, locoregional blocks, and epidural analgesia/anesthetics in this patient population as superior to multimodal analgesia. Of note, a previous guideline was published with a different recommendation on epidural catheters, suggesting their use.[63] This conclusion resulted from two studies not included here.[64,65] One included data from a conference abstract that was not subsequently published and thus did not meet inclusion criteria for this review. The second primarily examined inflammatory markers but also noted a decrease in numeric pain scores on days 1 and 3 after catheter placement; this study could not be meta-analyzed because standard deviation was not provided. Data review did not demonstrate convincing data for the PICO questions; thus, our recommendation differs from the previous one. These recommendations should not be interpreted as meaning epidural analgesia, and other locoregional techniques should not be used, nor should they be interpreted as a recommendation to increase opioid use. Rather, we advocate a multifaceted pain management strategy based on multimodal analgesia and other techniques according to provider judgment and institutional resources.

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AUTHORSHIP

All authors contributed in the development of PICO questions, voting on outcomes, voting on recommendations, critical review, and editing of manuscript. K.M. contributed in the meta-analysis, formulation of recommendations with group supervision, and writing manuscript.
S.D.S. contributed in the writing manuscript and review of articles. G.T. contributed in the writing manuscript and review of articles. B.K., K.L.H., K.L.K., R.B., K.S., A.M.S., and Z.M.B. contributed in the review of articles. H.K., M.C., S.K.A., J.J.C., E.P.H., and G.K. contributed in the development of recommendations, critical overview of K.M.'s meta-analysis, and formulation of recommendations.

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DISCLOSURE

S.D.S. and Z.M.B. are educational consultants for Zimmer Biomet Inc. A.M.S. is an education consultant for Globus Medical, Inc. and for Synthes USA Products LLC. The rest of the authors declare no conflicts of interest.

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